

1 Integer finite-chain input

This placeholder supplies the preceding-section reference used by the inputtable checkpoint fragment.

2 Generated five-term relations and the finite-boundary bridge

Author: ChatGPT. The valuation-contact construction of Section 1 provides a concrete finite-support exterior surface for every positive reduced rational cell and an analytic estimate for every finite chain satisfying an exact surface boundary. What remained was to derive that boundary from the relation generated by five-term moves. We supply that derivation here. We first work in a free divisor-symbol module, then pass to the smaller relation generated only by positively realized rational moves. The argument is the divisor shadow of the standard pre-Bloch complex [1, 2, 3]; no K -theoretic or abc estimate is imported from that theory.

2.1 The free symbol chain and its boundary

Let

$$\mathcal{D} = \mathbb{N} \rightarrow_0 \mathbb{Z}, \quad \mathcal{E} = (\mathbb{N} \times \mathbb{N}) \rightarrow_0 \mathbb{Z},$$

and retain the ordered-coordinate exterior product

$$(A \wedge B)_{p,q} = A_p B_q - A_q B_p.$$

Write $\mathcal{S} = \mathcal{D} \times \mathcal{D}$ for the set of divisor symbols and let

$$\mathcal{C}_0 = \mathbb{Z}^{(\mathcal{S})}$$

be the free integral module on those symbols. Its basis element associated with (A, B) is denoted by $[A, B]$. Define a linear map

$$\partial : \mathcal{C}_0 \longrightarrow \mathcal{E}, \quad \partial[A, B] = A \wedge B. \tag{1}$$

For $X, U, Y, V, Z \in \mathcal{D}$, define the formal five-term chain

$$\begin{aligned} \mathfrak{t}(X, U, Y, V, Z) = & [X, U] - [Y, V] + [Y - X, Z - X] \\ & - [Y + U - X - V, Z - X - V] + [U - V, Z - V]. \end{aligned} \tag{2}$$

Theorem 2.1 (Five-term generators are boundary cycles). *For all $X, U, Y, V, Z \in \mathcal{D}$,*

$$\partial \mathfrak{t}(X, U, Y, V, Z) = 0.$$

Proof. By linearity and (1), the claimed boundary is

$$\begin{aligned} & X \wedge U - Y \wedge V \\ & \quad + (Y - X) \wedge (Z - X) \\ & \quad - (Y + U - X - V) \wedge (Z - X - V) \\ & \quad + (U - V) \wedge (Z - V). \end{aligned}$$

Expand each wedge bilinearly. Every elementary wedge between two members of $\{X, U, Y, V, Z\}$ occurs with opposite coefficients and cancels. This is also the finite-support lift of the earlier coefficientwise five-term identity. $\square$

Let

$$\mathcal{R}_5 = \text{span}_{\mathbb{Z}} \{ \mathfrak{t}(X, U, Y, V, Z) : X, U, Y, V, Z \in \mathcal{D} \} \subseteq \mathcal{C}_0. \tag{3}$$

Theorem 2.2 (Generated relation-to-boundary bridge). *One has*

$$\mathcal{R}_5 \subseteq \ker \partial.$$

Consequently, if

$$c \sim_5 d \iff c - d \in \mathcal{R}_5,$$

then $\partial c = \partial d$.

Proof. The kernel of ∂ is an integral submodule. By Theorem 2.1, it contains every generator of the span in (3); hence it contains the span. If $c - d \in \mathcal{R}_5$, then linearity gives $\partial c - \partial d = \partial(c - d) = 0$. $\square$

This proof includes arbitrary finite sums, negatives, and integer multiples. In particular, the exact boundary is a consequence of a generation certificate and is not included among its premises.

2.2 The positive rational subrelation

For a positive reduced cell

$$P = (a, b, c), \quad a + b = c, \quad \gcd(a, b) = 1,$$

define its divisor symbol by

$$\sigma(P) = (d(a) - d(c), d(b) - d(c)). \quad (4)$$

Then

$$\partial[\sigma(P)] = (d(a) - d(c)) \wedge (d(b) - d(c)) = \Omega(P). \quad (5)$$

A positive realization of a five-term generator consists of positive reduced cells $P_1, \dots, P_5$ and divisor data X, U, Y, V, Z for which

$$\begin{array}{c|l} P_1 & (X, U) \\ P_2 & (Y, V) \\ P_3 & (Y - X, Z - X) \\ P_4 & (Y + U - X - V, Z - X - V) \\ P_5 & (U - V, Z - V) \end{array} \quad (6)$$

are the five individual identities for $\sigma(P_i)$. These local symbol identities contain no signed boundary assertion. Let $\mathcal{R}_5^+$ be the integral span of the signed chains

$$[\sigma(P_1)] - [\sigma(P_2)] + [\sigma(P_3)] - [\sigma(P_4)] + [\sigma(P_5)] \quad (7)$$

over all positive realizations.

Theorem 2.3 (Positive generated boundary). *There are inclusions*

$$\mathcal{R}_5^+ \subseteq \mathcal{R}_5 \subseteq \ker \partial.$$

Thus every equivalence generated by actual positive five-term moves preserves the finite-support surface.

Proof. The five identities in (6) identify each chain (7) with the algebraic generator (2). Therefore each generator of $\mathcal{R}_5^+$ belongs to $\mathcal{R}_5$, and the first inclusion follows by taking spans. The second inclusion is Theorem 2.2. $\square$

Suppose now that a target cell P_0 and cells P_j with coefficients $n_j \in \mathbb{Z}$ satisfy the generation statement

$$[\sigma(P_0)] - \sum_j n_j [\sigma(P_j)] \in \mathcal{R}_5^+. \quad (8)$$

Theorem 2.3 and (5) derive

$$\Omega(P_0) = \sum_j n_j \Omega(P_j). \quad (9)$$

Applying the exact finite-chain theorem from the preceding section gives

$$\Phi(P_0) \leq \frac{1}{2} \left(\sum_j |n_j| (M(P_j) + V(P_j)) + \sum_j |n_j| R(P_j) \right). \quad (10)$$

Unlike the earlier conditional structure, (8) stores only membership in the relation generated by actual moves. Equation (9) is a theorem.

2.3 An infinite arithmetic family of positive moves

Let $0 < b < a < c$ and impose the five reducedness conditions

$$\begin{aligned} \gcd(a, c-a) &= \gcd(b, c-b) = \gcd(b, a-b) = 1, \\ \gcd(b(c-a), c(a-b)) &= 1, \\ \gcd(c-a, a-b) &= 1. \end{aligned} \quad (11)$$

Consider the five cells

	left	middle	total
P_1	a	$c-a$	c
P_2	b	$c-b$	c
P_3	b	$a-b$	a
P_4	$b(c-a)$	$c(a-b)$	$a(c-b)$
P_5	$c-a$	$a-b$	$c-b$

(12)

Every entry is positive and the hypotheses (11) make every row reduced. The fourth row is a cell because

$$b(c-a) + c(a-b) = a(c-b).$$

Proposition 2.4 (Common-denominator realization). *The cells (12) positively realize the five-term move with*

$$x = \frac{a}{c}, \quad y = \frac{b}{c},$$

and arguments

$$\frac{a}{c}, \quad \frac{b}{c}, \quad \frac{b}{a}, \quad \frac{b(c-a)}{a(c-b)}, \quad \frac{c-a}{c-b}. \quad (13)$$

Proof. Put

$$\begin{aligned} X &= d(a) - d(c), & U &= d(c-a) - d(c), \\ Y &= d(b) - d(c), & V &= d(c-b) - d(c), \\ Z &= d(a-b) - d(c). \end{aligned}$$

The symbols of the first two cells are (X, U) and (Y, V) by definition. Direct subtraction gives the third and fifth rows of (6). For the fourth row, multiplicativity of divisors gives

$$\begin{aligned} d(b(c-a)) - d(a(c-b)) &= Y + U - X - V, \\ d(c(a-b)) - d(a(c-b)) &= Z - X - V. \end{aligned}$$

Thus all five symbol identities hold, and Theorem 2.3 supplies the surface relation. $\square$

For every odd $c \geq 3$, the choice $a = 2$, $b = 1$ satisfies (11). Indeed, the only nontrivial checks reduce to

$$\gcd(2, c - 2) = 1, \quad \gcd(c - 2, c) = \gcd(c - 2, 2) = 1.$$

This gives an infinite parameterized family of actual positive generators.

2.4 A full-premise counterexample to literal chain cancellation

The generated quotient is essential. Define the augmentation

$$\varepsilon : \mathcal{C}_0 \longrightarrow \mathbb{Z}, \quad \varepsilon \left(\sum_s n_s [s] \right) = \sum_s n_s.$$

For every input one has

$$\varepsilon(\mathfrak{r}(X, U, Y, V, Z)) = 1 - 1 + 1 - 1 + 1 = 1. \quad (14)$$

Proposition 2.5 (A generator is never the zero free chain). *For every $X, U, Y, V, Z \in \mathcal{D}$,*

$$\mathfrak{r}(X, U, Y, V, Z) \neq 0.$$

In particular, the boundary map ∂ is not injective.

Proof. If the free chain were zero, its augmentation would be zero, contradicting (14). Its boundary is nevertheless zero by Theorem 2.1, so ∂ has a nonzero kernel. $\square$

This refutes the exact shortcut asserting that the five signed basis chains cancel before quotienting. It does not refute the five-term route. The correct statement is the kernel inclusion of Theorem 2.2.

The failure persists with all positivity and reducedness premises. Taking

$$x = \frac{1}{2}, \quad y = \frac{1}{6}$$

gives the five distinct arguments

$$\frac{1}{2}, \quad \frac{1}{6}, \quad \frac{1}{3}, \quad \frac{1}{5}, \quad \frac{3}{5}$$

and reduced cell triples

$$(1, 1, 2), \quad (1, 5, 6), \quad (1, 2, 3), \quad (1, 4, 5), \quad (3, 2, 5).$$

Their signed surface boundary is zero, whereas their signed free chain is nonzero; its coefficient at the $1/2$ symbol is one.

2.5 Formal boundary and remaining analytic gate

The Lean module `SteinbergFiveTermBoundaryBridge20260903` formalizes Theorems 2.1, 2.2, and 2.3, the generated filling implication (10), Proposition 2.4, the odd-denominator family, and Proposition 2.5. It also formalizes the five-distinct positive example. Its axiom audit uses only `propext`, `Classical.choice`, and `Quot.sound`.

The algebraic part of S-I4C is therefore closed: a chain generated by actual positive five-term moves has the required finite exterior boundary. The route remains open at the harder existence and analytic stages. For an arbitrary target cell, one must still construct a finite

positive generated filling satisfying useful bounds on chain length, coefficients, prime support, and the two costs on the right of (10). More precisely, put

$$\mathcal{Q}(\Gamma) = \sum_j |n_j|(M(P_j) + V(P_j)), \quad \mathcal{R}(\Gamma) = \sum_j |n_j|R(P_j).$$

For every $\epsilon > 0$, Gate VF still requires constants K_ϵ, L_ϵ and a generated filling for each primitive target of height $H = \log c$ and radical logarithm $\rho = \log \text{rad}(abc)$ such that

$$\mathcal{R}(\Gamma) \leq \epsilon \mathcal{Q}(\Gamma) + K_\epsilon H, \quad \mathcal{Q}(\Gamma) \leq 2H\rho + L_\epsilon H.$$

Neither estimate nor such a uniform construction is proved here, and this section proves neither the *abc* conjecture nor its negation.

References

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